

Circuit Lab

MIT Invitational 2026 — January 24, 2026

Written by:

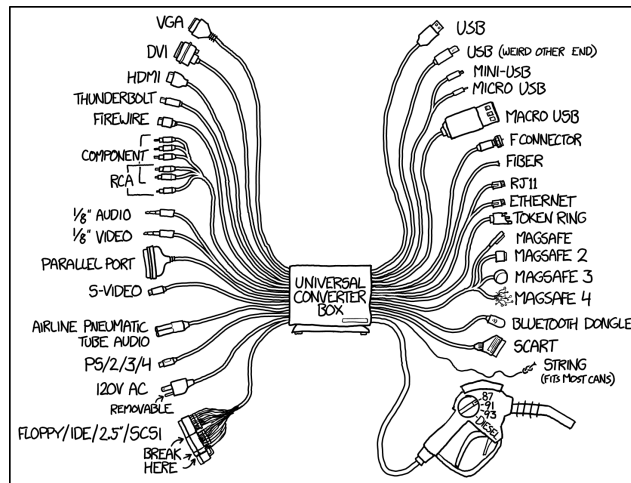
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Answer Sheet



Section	Page	Tie Break	Points	Score
0. Multiple Choice	1	TB7	50	
1. Ducks	2	TB6	20	
2. Dielectric Dilemma	3	TB3	15	
	4		15	
3. Closely Relayed	5	TB2	13	
	6		12	
4. Lightly Extracting Data	7	TB5	12	
	8		8	
	9		20	
5. The I(V) Lab	10	TB1	19	
	11		21	
6. Gray Box	12	TB4	12	
	13		28	

FINAL SCORE:

RANKING:

School/Team: _____ Team #: _____

0 Multiple Choice [50 points]

- Instructions:
- Bubble in your team number with leading zeros and without the "C". For example, if you are C73, bubble in 073.
 - Bubble in your answers the 25 multiple choice questions.
 - Make sure you completely darken the bubble corresponding to your response.
 - **Do not use** this page as scratch paper.

Team Name

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1 (A) (B) (C) (D) 16 (A) (B) (C) (D)

2 (A) (B) (C) (D) 17 (A) (B) (C) (D)

3 (A) (B) (C) (D) 18 (A) (B) (C) (D)

4 (A) (B) (C) (D) 19 (A) (B) (C) (D)

5 (A) (B) (C) (D) 20 (A) (B) (C) (D)

6 (A) (B) (C) (D) 21 (A) (B) (C) (D)

7 (A) (B) (C) (D) 22 (A) (B) (C) (D)

8 (A) (B) (C) (D) 23 (A) (B) (C) (D)

9 (A) (B) (C) (D) 24 (A) (B) (C) (D)

10 (A) (B) (C) (D) 25 (A) (B) (C) (D)

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15 (A) (B) (C) (D)

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MIT 2026 Circuit Lab (9323)

1 Ducks [20 points]

26. (a) [3 points] Express the output voltage v_{out} in terms of the voltages V_1, V_2, V_3, V_4 .

- (b) [3 points] How many discrete values of v_{out} can this circuit emulate? List out configurations of V_1, V_2, V_3, V_4 that produce the lowest four values of v_{out} this circuit can emulate.

- (c) [2 points] What is the smallest voltage change Δv_{out} that this circuit can create?

- (d) [8 points] Express (i) $v_{\text{out}}(V_1, \dots, V_n)$, (ii) the number of values v_{out} can take on, and (iii) the minimum change in voltage Δv_{out} .

- (e) [4 points] What is the change in output voltage $v_1 - v_0$ due to the addition of the $10R$ load? Give your answer as a fraction times V .

2 Dielectric Dilemma [30 points]

27. (a) [4 points] What is the overall capacitance this method yields? Simplify your answer to $\epsilon_0 \frac{A}{d}$ times some function of κ .

- (b) [4 points] What is the overall capacitance this method yields? Simplify your answer to $\epsilon_0 \frac{A}{d}$ times some function of κ .

- (c) [4 points] You should not get the same result. Why is this?

- (d) [3 points] What is the other value of κ that makes (a) and (b) equal? _____

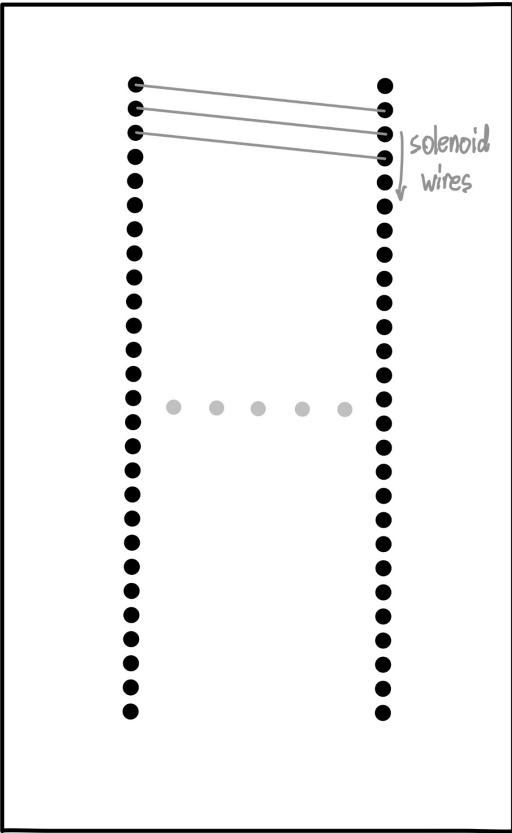
28. (a) [5 points] Find $U(\theta)$ in terms of E_0 , d , q , and θ .

(b) [5 points] For what values of $k_B T$ is the probability the crystal is in the maximal scattering orientation less than 1%? Write your answer as an inequality between $k_B T$ and some combination of E_0 , d , and q times a decimal (to 3 significant figures).

(c) [5 points] What is the expected total amount of energy needed to align all these crystals in the same field $\mathbf{E} = E_0 \hat{\mathbf{x}}$, in terms of E_0 , q , d , and N ?

3 Closely Relayted [25 points]

29. (a) [2 points] Indicate whether the current must flow in the **positive or negative** direction (as indicated in the diagram) to make the arm touch the upper contact. _____
- (b) [4 points] Draw **five magnetic field lines** showing the magnetic field produced by the solenoid in this situation (where the arm touches the upper contact) **that cross through the five dots below**. The lines that cross through the furthest left and furthest right dots also cross through the wires of the solenoid.



This is the cross-section view of the solenoid from the same perspective as the diagram on the questions sheet. The wires come out of the page from the dots on the left, wrap to the right and down, then go back into the page.

30. (a) [3 points] What is the minimum voltage $V_{\min}$ applied between terminals A and B required to make the contacts touch?

- (b) [4 points] Fill in the table below, assuming there is no resistance within the boxes.

I_1	I_2	O_1
0V	0V	
5V	0V	
0V	5V	
5V	5V	

- (c) i. [5 points] Sketch the capacitor's voltage over time on the graph provided. Make sure to note critical voltage values. No labels are necessary for the time axis, but you should draw with a reasonable time scale.



- ii. [4 points] What is the graph's period?

- iii. [3 points] What value of C is needed to produce the note?

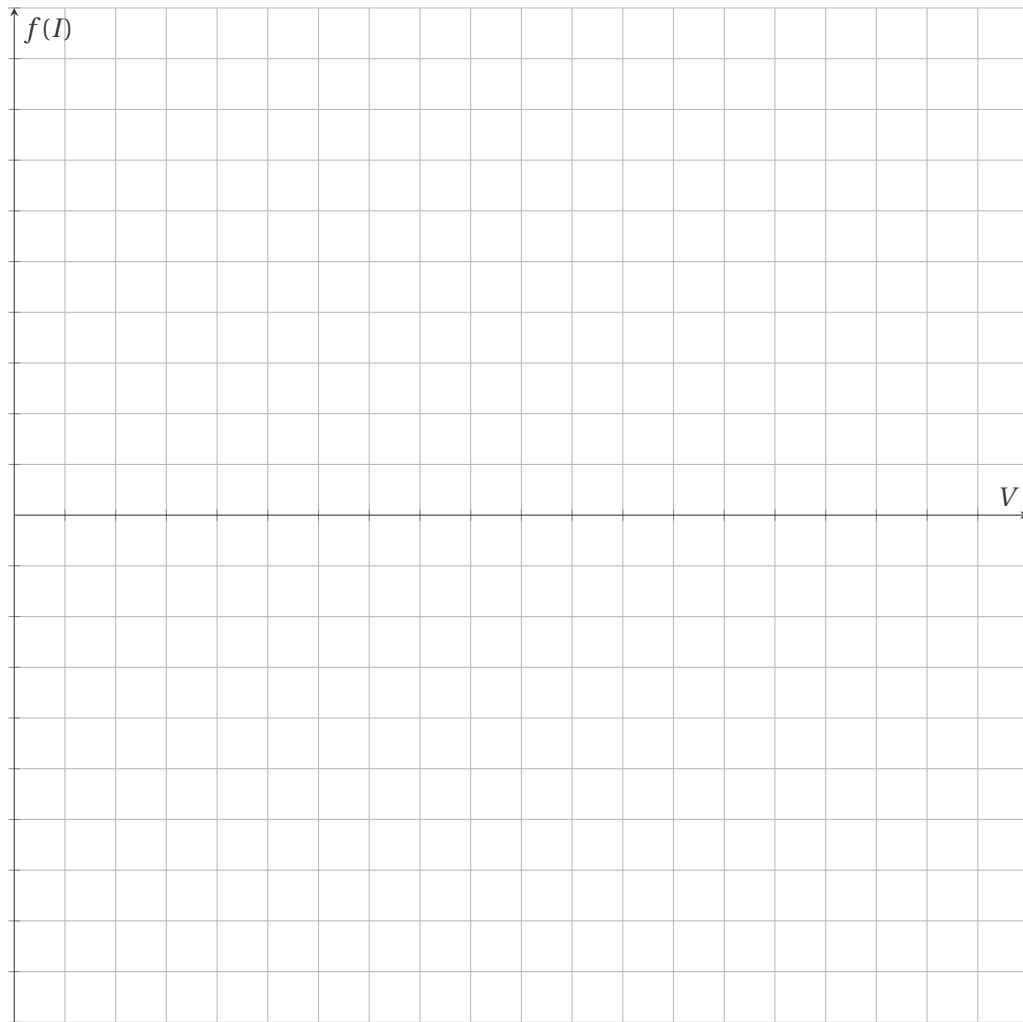
4 Lightly Extracting Data [40 points]

31. (a) [3 points] Find the value of b for this diode.

- (b) [4 points] Rearrange the diode equation to create an expression of the form $f(I) = aV + m_1$, where $f(I)$ and m_1 are expressions that do not depend on V you must find.

- (c) [5 points] Create and fill out a table with the values of $f(I)$ for every value V you will plot.

- (d) [6 points] Graph your data on the provided graph, with V on the x-axis and $f(I)$ on the y-axis. Estimate the slope by drawing a line of best fit. What is a for this diode, and what is its forward voltage?



- (e) [2 points] Is the forward voltage closer to that of a germanium or silicon diode? _____

32. (a) [3 points] When the temperature of the diode increases, does its forward voltage increase or decrease? Explain why, on a charge-carrier level.

- (b) [3 points] When temperature increases, does a decrease more or does b decrease more?

- (c) [2 points] Does the magnitude of reverse current increase or decrease with increasing temperature?

- (d) [3 points] What happens to electrons in the depletion zone of the diode to cause it to divert from this equation? Assume the depletion zone is not very thin.

- (e) [3 points] Explain why silicon's indirect band gap makes it difficult to produce visible light in energy transitions.

- (f) [3 points] Can the LED achieve 100% efficiency in converting the light into electricity? Explain.

- (g) [3 points] Where could the extra energy have come from?

5 The I(V) Lab [40 points]

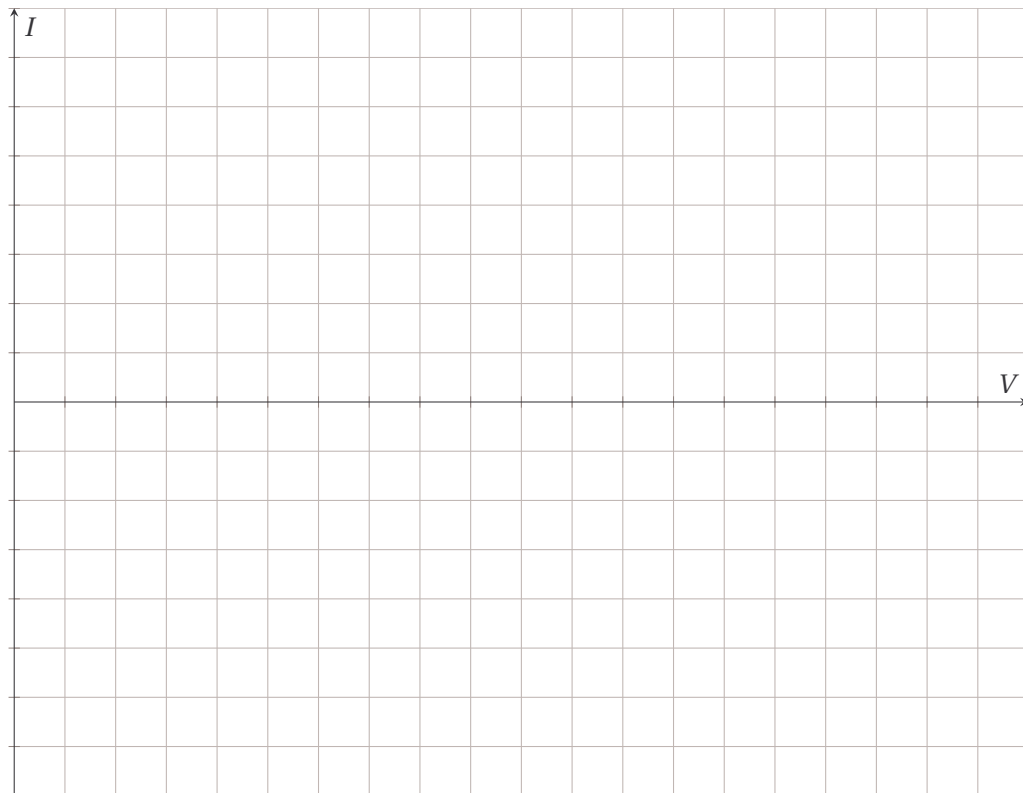
Read the safety guidelines on your test booklet before beginning.

33. (a) [15 points] Fill out 11 measurements in the table below, reasonably spanning the full range of voltages V_p you can produce across POT.

V_2	V_p

- (b) [4 points] If you measure a voltage V_2 across R_2 and V_p across POT (as described in the previous part), what is the voltage applied and current flowing through the component?

- (c) [8 points] Plot the voltage applied to the circuit versus the current flowing through it in the direction of the applied voltage. If you could not do the previous part, graph V_p on the x-axis and V_2 on the y-axis. You must draw and label an appropriate scale on the axes of the graph. You may draw a curve to fit the data, but make sure your plotted points are still visible.



- (d) [4 points] Using your graph, what is the equivalent resistance of the circuit while the applied voltage is low?

- (e) [4 points] Circle the two distinct regions in the graph you drew and explain why the graph has two distinct regions.

- (f) [5 points] Determine the op-amp's maximum output voltage from your graph. Show how you got this value.

6 Gray Box [40 points]

Read the safety guidelines on your test booklet before beginning.

34. (a) [12 points] Determine the identity of components **A**, **B**, and **C**. Show your work and clearly box your answer.

- (b) [8 points] Determine the resistance of the resistor (in the gray box) in $k\Omega$. Write down the quantities you measured and how you measured them.

- (c) [10 points] Determine the capacitance of the capacitor in μF . Write down the quantities you measured and how you measured them.

- (d) [10 points] Determine the forward drop of the diode in V. Write down the quantities you measured and how you measured them.